3.3 Solving Quadratic Equations

Name _____ Period ____ Date _____

Algebra II 3.3 (30-minute plan) · Student sheet

I can…

- solve quadratic equations using square roots.
- solve quadratic equations by factoring and by graphing.
- choose a method and use zeros to solve real-life problems.

Vocabulary · fill in the missing word

Quadratic equation — an equation that can be _____ $ax^2 + bx + c = 0$, $a \neq 0$

Root of an equation — a _____

Zero of a function — an _____ where $f(x) = 0$

Square roots — if $x^2 =$ _____, then $x = \pm\sqrt{d}$

Part 1 · Notes and examples (fill in each note, then work the example)

Square roots: If $x^2 = d$ $(d > 0)$, then $x =$ _____.

Example 1

Solve each equation.

a. $2x^2 - 50 = 0$ b. $(x - 3)^2 = 7$

Factoring: Before factoring, the equation must equal _____.

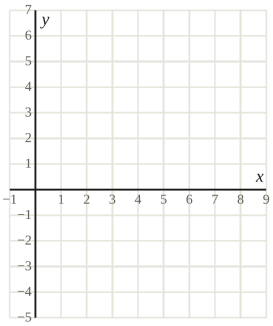
Example 2

Solve $x^2 + 3x = 18$.

On a graph: On a graph, the solutions are the _____.

Example 3

Find the zeros of $f(x) = x^2 - 8x + 12$. Check with the graph.



Part 2 · Partner practice

Partner practice

Solve $3x^2 = 48$.